* character table ($\omega = e^{2\pi i/3}$)

| $T(c)$ | $1(1)$ | $2_{001}(3)$ | $3^{+}_{111}(4)$ | $3^{-}_{111}(4)$ |
|---|---|---|---|---|
| $A$ | 1 | 1 | 1 | 1 |
| $E^{(a)}$ | 1 | 1 | $\omega^*$ | $\omega$ |
| $E^{(b)}$ | 1 | 1 | $\omega$ | $\omega^*$ |
| $T$ | 3 | $-1$ | 0 | 0 |

* polar $\leftrightarrow$ axial conversion

$A$ ($A$) $E^{(a)}$ ($E^{(a)}$) $E^{(b)}$ ($E^{(b)}$) $T$ ($T$)

* symmetric product

| | $A$ | $E^{(a)}$ | $E^{(b)}$ | $T$ |
|---|---|---|---|---|
| $A$ | $A$ | $E^{(a)}$ | $E^{(b)}$ | $T$ |
| $E^{(a)}$ | | $E^{(b)}$ | $A$ | $T$ |
| $E^{(b)}$ | | | $E^{(a)}$ | $T$ |
| $T$ | | | | $A + E^{(a)} + E^{(b)} + T$ |

* anti-symmetric product

| $A$ | $E^{(a)}$ | $E^{(b)}$ | $T$ |
|---|---|---|---|
| $-$ | $-$ | $-$ | $T$ |